Experiment No. 3 – Explore Version Control Systems

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Aim

To explore the Git Version Control System and practice source code management using GitHub.

Objectives

- To understand the concepts of Version Control Systems (VCS).
- To learn Git fundamentals and repository management.
- To practice source code management using GitHub.
- To perform basic Git operations such as configuration, status checking, staging, committing, and pushing code.

Theory

A **Version Control System (VCS)** is a software tool that helps developers track and manage changes to source code over time. It enables collaboration, maintains a history of modifications, and allows reverting to previous versions if needed.

Git is a distributed version control system widely used for software development. **GitHub** is a cloud-based platform that hosts Git repositories and facilitates collaboration among developers.

Benefits of Version Control

- Tracks changes in source code.
- Enables collaboration among developers.
- Maintains project history. Supports branching and merging.
- Allows rollback to previous versions.
- Facilitates Continuous Integration and Continuous Deployment (CI/CD).

Software Requirements

- Git
- GitHub Account
- Internet Connection
- Source Code from Experiment No. 2

Procedure

Step 1: Install Git

Download and install Git from:

Git Official Website

Verify installation:

Go to command prompt=> run as administrator

git --version

Step 2: Configure Git

Configure username and email:

git config --global user.name "Your Name"

git config --global user.email "ajitpatil@gmail.com"

Verify configuration:

git config --list

Step 3: Create Local Repository

Create a project folder and initialize Git:

mkdir DevOpsLab

cd DevOpsLab

git init

Output:

Initialized empty Git repository

Step 4: Add Application Source Code

Copy the Python program developed in Experiment No. 2 into the repository folder.

Check repository status:

git status

Step 5: Stage Files

Add files to the staging area:

git add .

Check status again:

git status

Step 6: Commit Changes

Create a commit:

git commit -m "Initial Commit"

View commit history:

git log

Step 7: Create GitHub Repository

1. Login to GitHub.
2. Click **New Repository**.
3. Enter repository name: **DevOpsLab**.
4. Create repository.

Step 8: Connect Local Repository with GitHub

git remote add origin https://github.com/username/DevOpsLab.git

Verify remote repository:

git remote -v

Step 9: Push Source Code to GitHub

git branch -M main

git push -u origin main

to resolve proxy issue

```
C:\DevOpsLab_C18>git config --global http.proxy http://172.27.100.2:3128
C:\DevOpsLab_C18>git config --global https.proxy http://172.27.100.2:3128
C:\DevOpsLab_C18>git config --global --get http.proxy http://172.27.100.2:3128
C:\DevOpsLab_C18>git push -u origin main
```

1. `git config --global http.proxy [http://172.27.100.2:3128](http://172.27.100.2:3128)`
   - Configures Git globally to route standard HTTP requests through a proxy server located at IP address `172.27.100.2` on port `3128`.
2. `git config --global https.proxy [http://172.27.100.2:3128](http://172.27.100.2:3128)`
   - Configures Git globally to route secure HTTPS requests through the same proxy server.
3. `git config --global --get http.proxy`
   - Retrieves and displays the current globally configured HTTP proxy URL (`[http://172.27.100.2:3128](http://172.27.100.2:3128)`) to confirm the configuration was saved correctly.
4. `git push -u origin main`
   - Pushes the local `main` branch commits to the remote repository named `origin` and sets `origin/main` as the default upstream tracking branch.

Common Git Commands

Command	Description
git init	**Initialize repository**
git status	**Show repository status**
git add	**Add files to staging area**
git commit	**Save changes**
git log	**View commit history**
git clone	**Clone repository**
git push	**Upload changes**
git pull	**Download updates**
git branch	**List branches**
git checkout	**Switch branches**
git merge	Merge branches
git remote -v	Show remote repositories

Sample Output

$ git status

On branch main

nothing to commit, working tree clean

Architecture of Git

Working Directory

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Staging Area

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Local Repository

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Remote Repository (GitHub)